KIWi – KI-Wissensassistenz

AI-Powered Knowledge Assistance for the THWS CampusPortal

Problem

- University information is distributed across different sources (e.g., CampusPortal, google and manuals).
- Finding relevant and up-to-date answers is time-consuming.
- Traditional search can struggle with natural-language questions and process-related context.
- University information requires privacy-conscious processing.

Objectives

- Enable natural-language question answering about university processes.
- Retrieve relevant information from approved knowledge sources.
- Generate context-aware answers using RAG.
- Provide transparent answers with supporting source references and URLs.
- Evaluate retrieval quality, answer correctness and response time.

Methodology

Data Ingestion
Available university knowledge records are collected from CSV files, combining synthetic data and publicly available THWS information.

Data Preprocessing
Questions, answers, key facts, evidence and source details are cleaned and prepared for further processing.

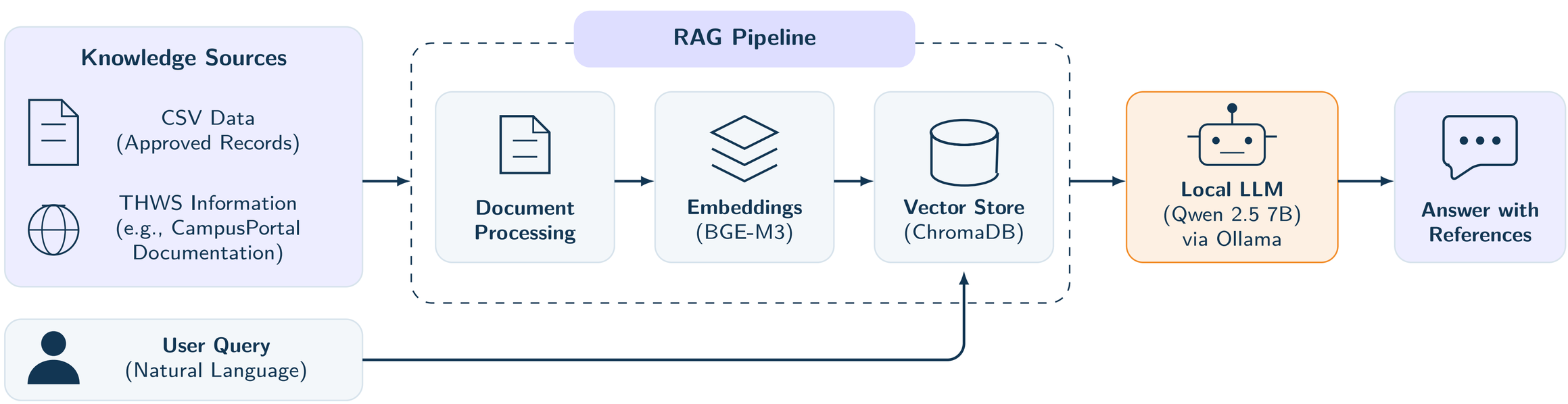
Text Chunking
Longer records are divided into smaller text passages while preserving relevant metadata and source information.

Embedding Generation
BGE-M3 converts the text passages into numerical vector representations for semantic search.

Information Retrieval
ChromaDB stores the embeddings and retrieves relevant passages based on the user's natural-language query.

Answer Generation
Qwen 2.5 7B Instruct, running locally through Ollama, generates context-aware answers using retrieved information and provides supporting source references.

Architecture



Knowledge ingestion prepares and indexes documents; at query time, relevant passages are retrieved and supplied to the local language model.

Deliverables / Expected Outcomes

Deliverables

- ✓ Working local RAG prototype.
- ✓ Development of a structured data ingestion workflow.
- ✓ Semantic knowledge storage using ChromaDB
- ✓ Interactive question-answering interface for natural-language queries.
- ✓ Source-supported answer generation.

Expected Outcomes

- ✓ Faster access to university information.
- ✓ More transparent and evidence-grounded answers.
- ✓ Reduced reliance on external AI services.
- ✓ A foundation for future CampusPortal integration and a web-based interface.

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